

Power Plant Engineering

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Course Outcomes

On completion of this course the student will be able to:

1. Get a general understanding of the scale and scope of hydroelectric power stations and also grasp the basics of diesel engine and gas turbine power plants.
2. Comprehend the complexity of the steam power plant after getting to know about the working of its different components and elements.
3. Understand the basic principle of the working of nuclear power plant and the various types of nuclear reactors with its safety aspects, especially with respect to the current scenario of national and international debates on the same.
4. Equip with the knowledge of the basics of various non-conventional energy producing methods .

Syllabus

Module I

Power plant economics – base load and peak load power plants -estimation of load – load curve – load factor – diversity factor – capacity factor – use factor – selection of units – number and size –scheduling operation – cost of energy – depreciation and replacement – economics of plant selection. Hydroelectric power plants – general Layout – types of dams penstock, draft tubes, surge tanks - power house equipments – site selection

Module II

Diesel engine power plant – Layout – Components of a diesel power plant – starting methods – Gas turbine – open and closed cycles – thermodynamics cycles– regeneration – reheating – intercooling – efficiency and performance of gas turbines. combustion chambers of gas turbines – cylindrical – annular and industrial type combustion chamber design– combustion efficiency – advantages and disadvantages Gas Turbine power plants – classification – elements of a GasTurbine power plant

Syllabus

Module Module III

Steam power plants General layout fuel handling systems types of furnaces stokers burning systems types of firing : stokers, pulverized coal burners and fluidized bed combustion power plant boilers, mountings and accessories dust and ash handling systems – draft and chimney calculations – condensers cooling systems - Environmental aspects of thermal power systems Nuclear power plants - Fundamentals of nuclear fission nuclear power plants reactors classification – components layout of simple plant – nuclear power safety and waste disposal.

Module IV

Non conventional energy sources solar radiation and its measurement Solar energy collectors Applications of solar energy Wind energy conversion – site selectio wind energy collectors – Energy from biomass - ocean energy possibilities and future scope – Ocean Thermal electric conversion (OTEC) – Tidal energy - geothermal energy- Magneto Hydro Dynamic (MHD) power– Fuel cells - thermo electric power - thermionic generation.

References:

1. Wakil, E. I., Power Plant Technology, McGraw Hill, (1985).
2. Nag, P. K., Power Plant Engineering, 3rd Edition, Tata McGraw Hill, (2007).
3. Morse, Power Plant Engineering, Van Nostrand Co., (1953).
4. Lee J. F., Power Station Engineering and Economy, Tata McGraw Hill, (1960).
5. Robert L. Loftness, Nuclear Power Plants, Van Nostrand, (1964).
6. Verma Mahesh, Power Plant Engineering, Metropolitan Book Co., (1976).
7. Rai G. D., Non Conventional Energy Sources, Khanna Publishers, (2004).
8. Cohen & Rogers, Gas Turbine Theory, 6th Edition, Prentice Hall, (2008).

Units

Temperature – K (Kelvin)

Pressure – Pa (N/m^2 or kg/ms^2) (Gauge or Absolute)

Flow rate - kg/s

Power - W (1000 W=1kW, 1000kW=1MW, 1000MW=1GW)

Unit of electric power- kWhr

Lower heating value (LHV) - kJ/kg

Higher heating value (HHV) – kJ/kg

Standard cubic feet (scf, commonly pronounced as “scuff”)

A standard cubic foot corresponds to 1 cubic foot of gas at 60 F (15.6 C) and 14.73 psia,

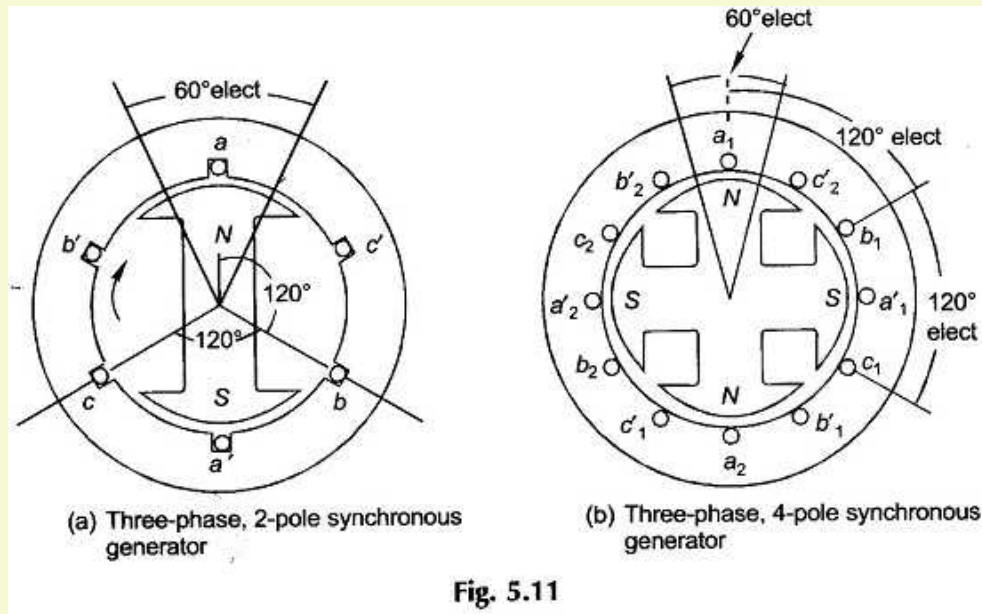
Normal cubic meters (Nm^3).

A normal cubic meter of gas corresponds to 1 cubic meter at 15 C at 101.325 kPa.

Basics

Electrics supply - 50 or 60 Hertz?

- 60 Hertz (Hz; e.g., in the USA)
- 50 Hz (e.g., in India Europe).
- In some countries (e.g., Saudi Arabia and Japan), both are present



Generator Frequency (f) = $PN/120$

Number of revolutions per minute of the engine (N)

Number of magnetic poles (P)

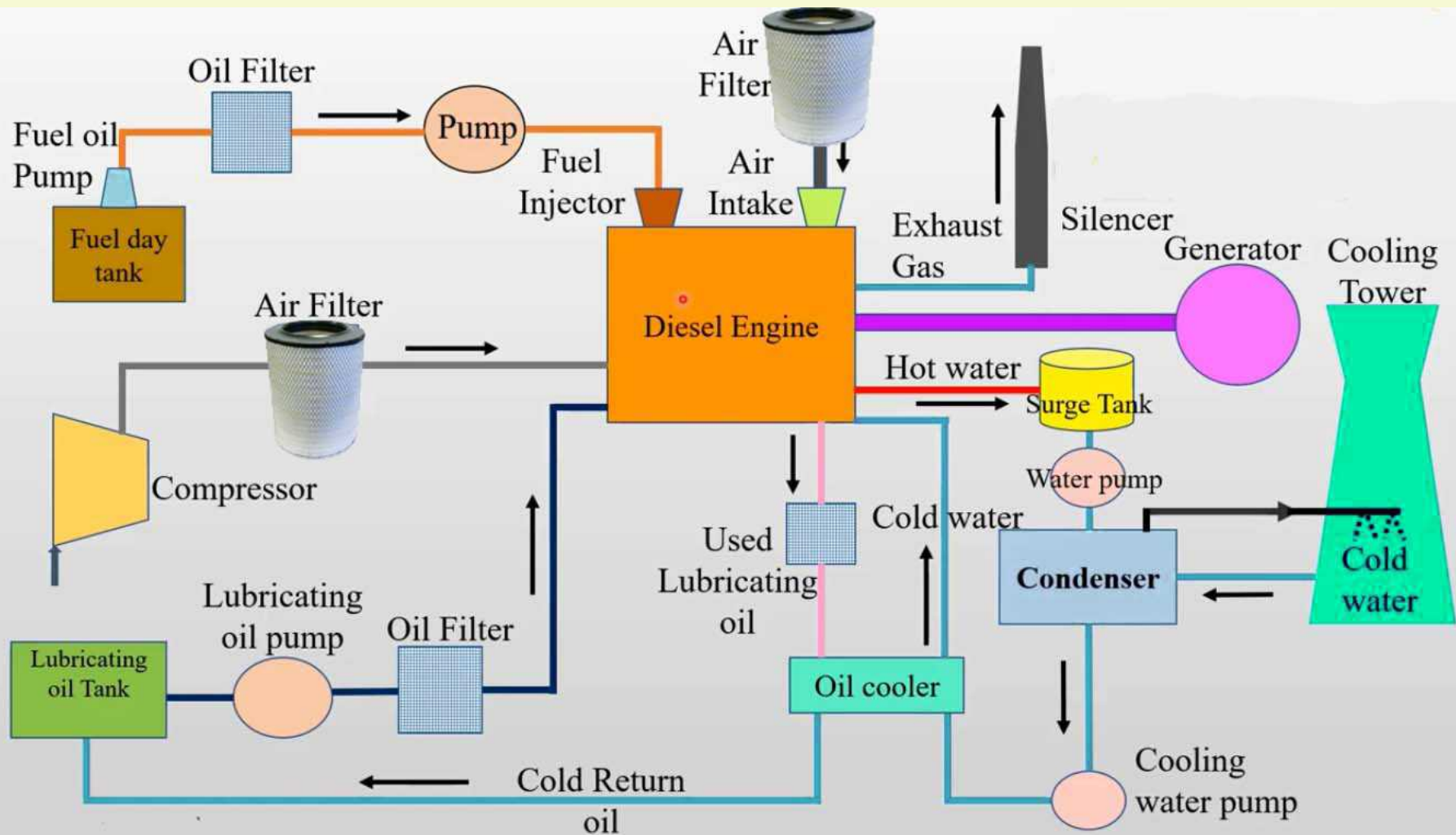
Diesel Power Plants

- Diesel power plants are – Diesel Engines Coupled to a Generator (2-50 MW)
- Diesel or Heavy Fuel Oil is usually used
- Applications
 - Peak Load plants
 - Mobile Plants
 - Standby Units
 - Emergency plants
 - Starting stations
 - Central Stations

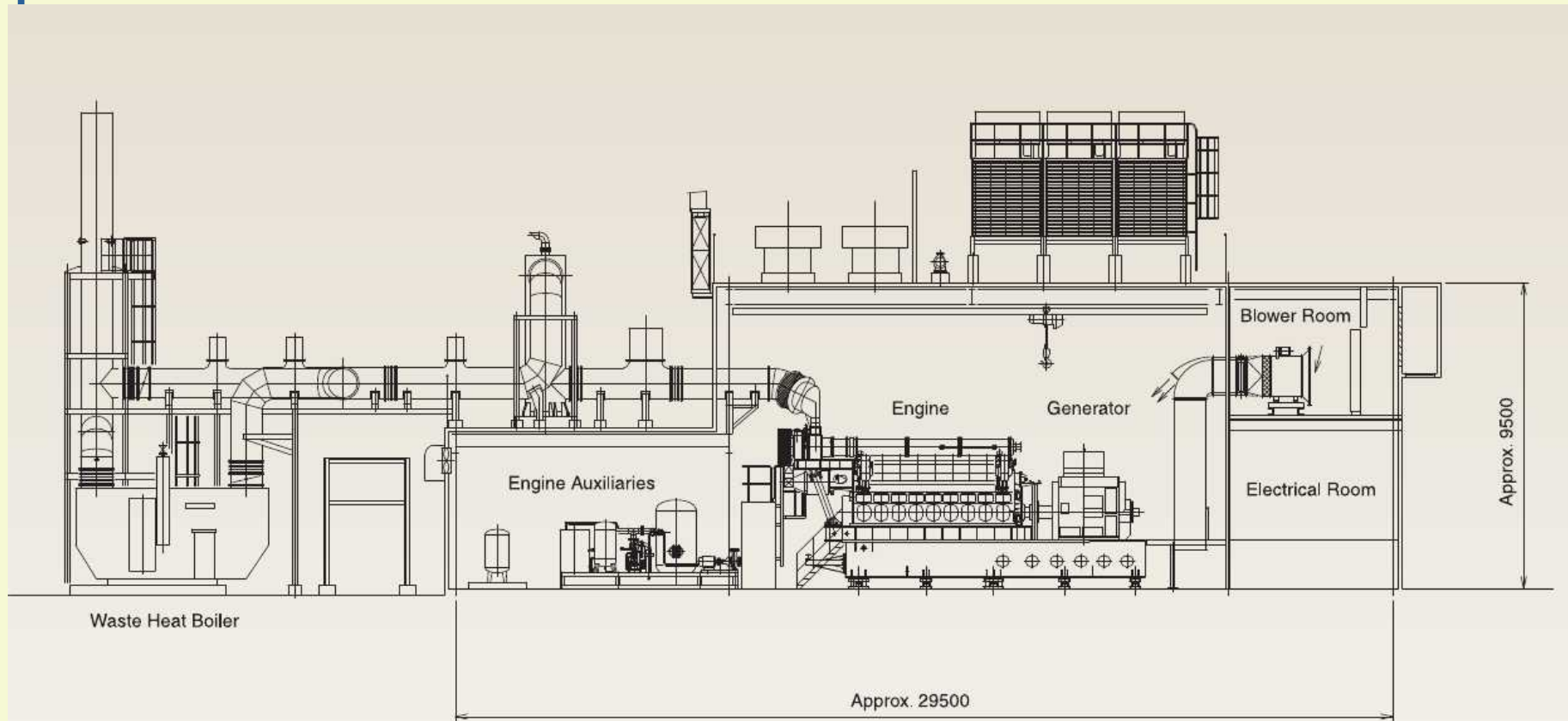
Diesel Power Plants Advantages and Disadvantages

- Advantages
 - Easy design of plant and installation
 - Available in different capacities
 - Response to load changes fast
 - Starting and stopping quick
 - Less capital required
 - High efficiency
 - Installation near load center
- Disadvantages
 - Operating cost high
 - Pollution and noise
 - High capacity limited

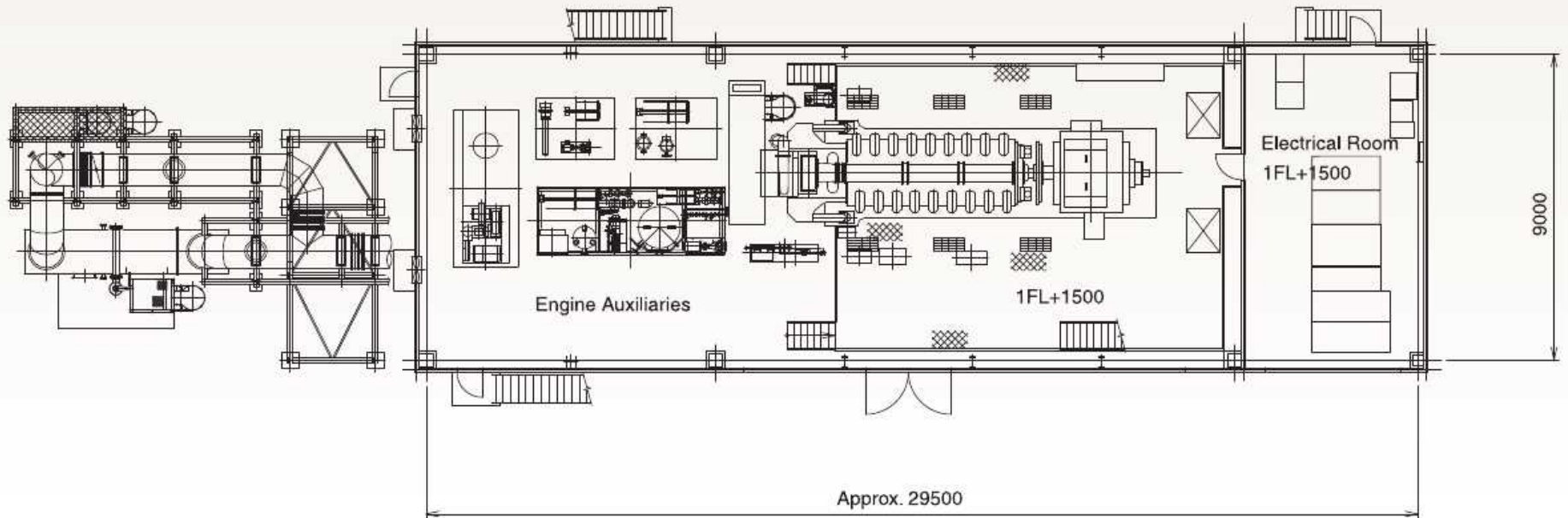
Diesel Power plant layout



Diesel Power plant layout



Diesel Power plant layout



Diesel Power plant manufacturers

Top Diesel power plant manufacturers

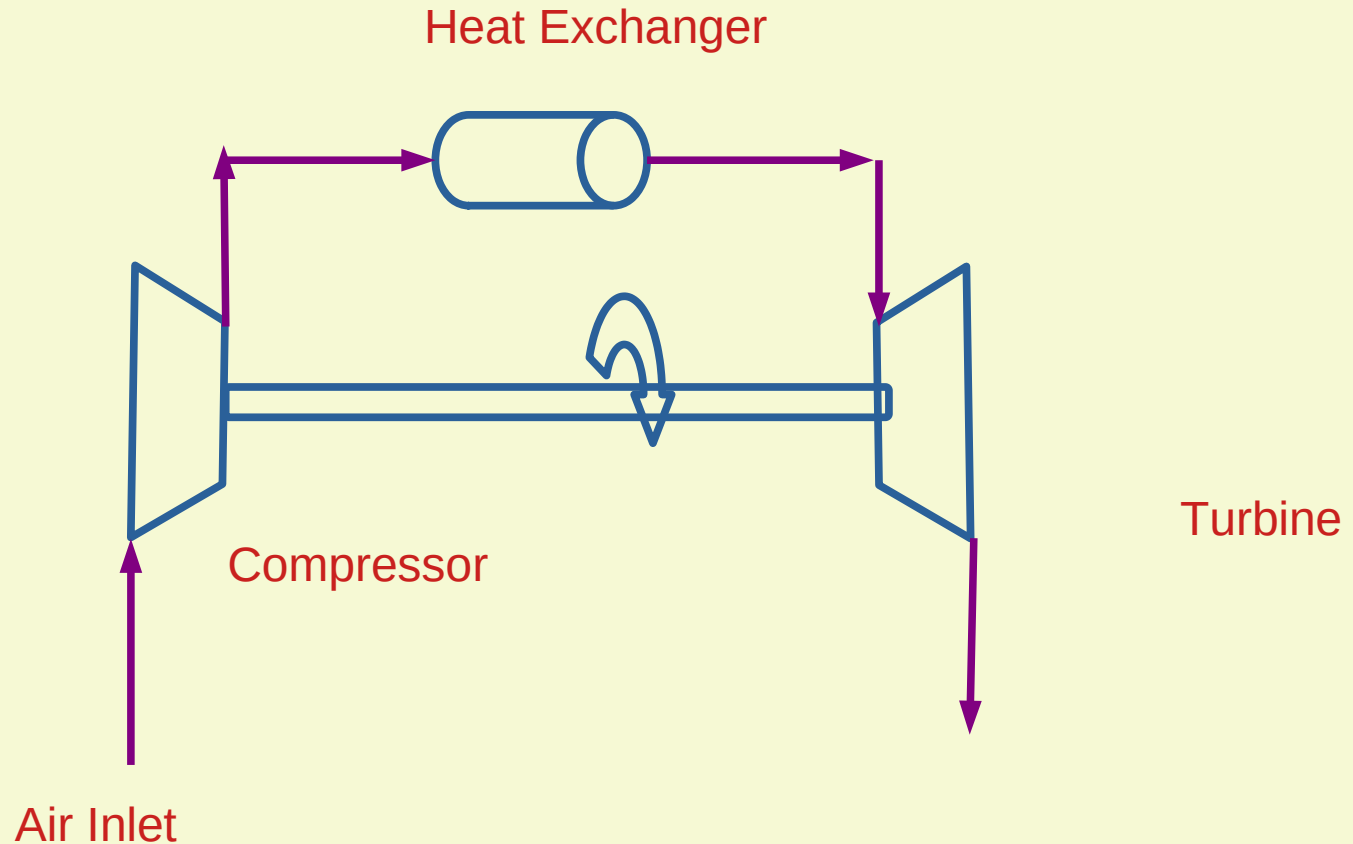
1. Atlas Copco
2. MAN Energy Solutions
3. Caterpillar Inc.
4. Cummins Inc.
5. Mitsubishi Heavy Industries Engine & Turbocharger, Ltd.
6. Kirloskar Oil Engines Limited
7. Mahindra Powerol

Diesel Power plant systems

1. Diesel Engine
2. Air intake system
3. Exhaust system
4. Fuel System
5. Cooling system
6. Lubrication system
7. Starting system

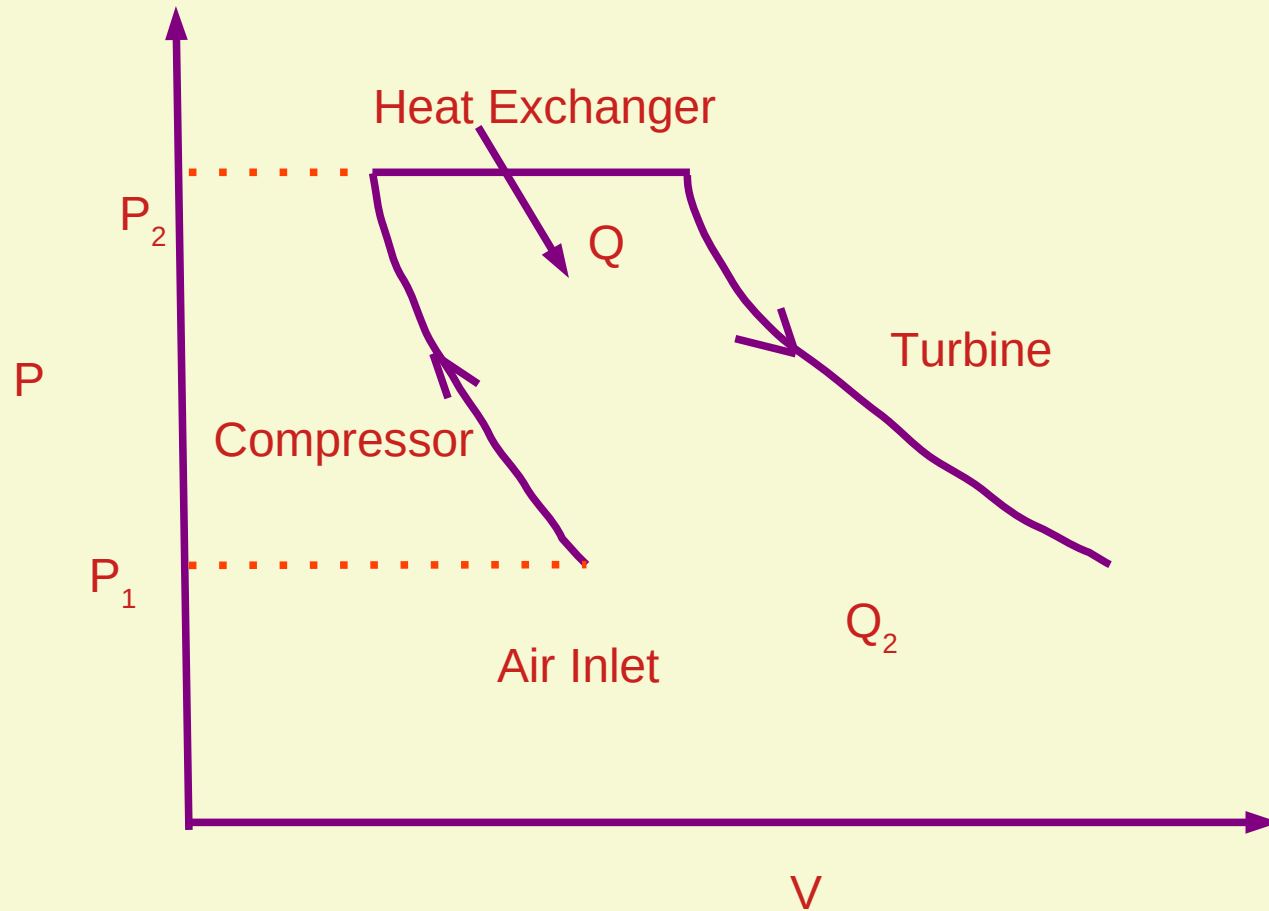
Gas Turbine Power Plants

1. Gas Turbine cycle (Open)



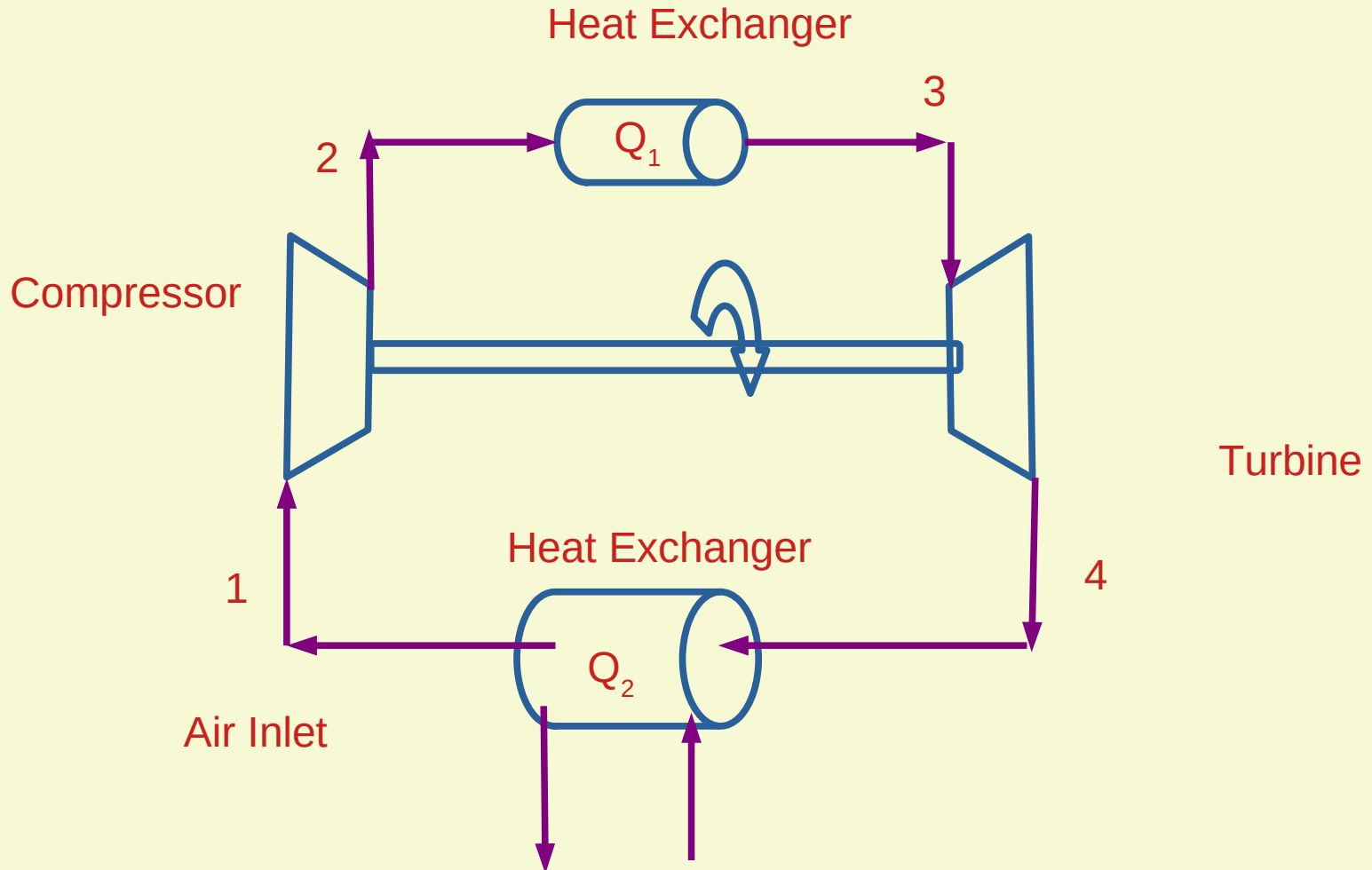
Gas Turbine Power Plants

1. Brayton cycle (Open)



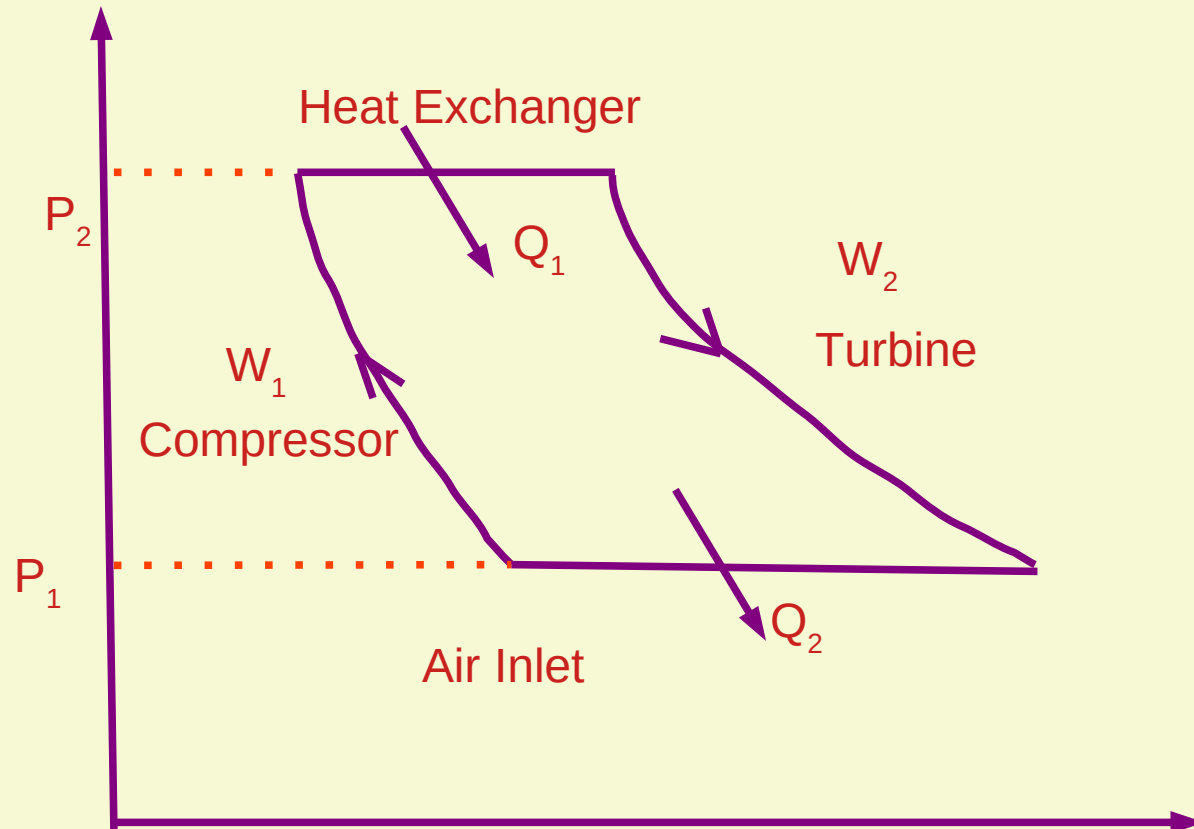
Gas Turbine Power Plants

1. Gas Turbine cycle (Closed)



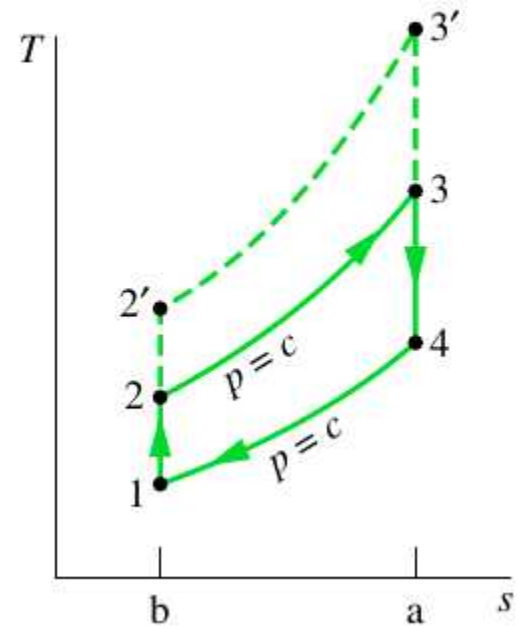
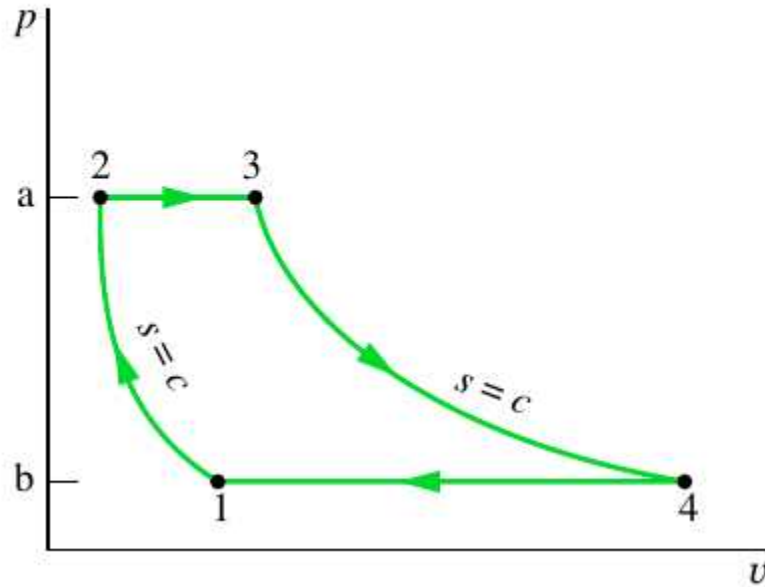
Gas Turbine Power Plants

Brayton cycle (Closed)



Gas Turbine Power Plants

1. Ideal Brayton cycle



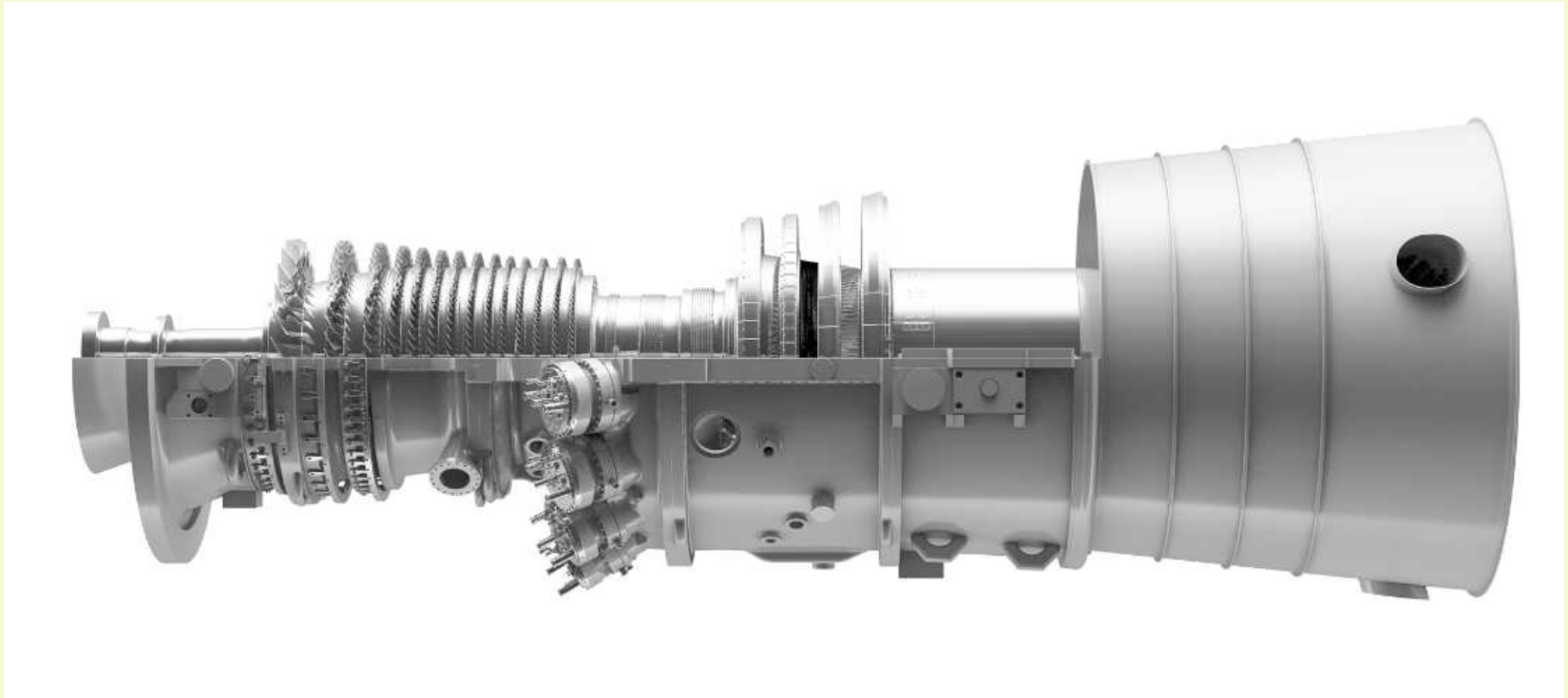
Gas Turbine Power plant systems

1. GE Gas turbine animation

<https://www.youtube.com/watch?v=x1aiMVFT8Bw>

<https://www.youtube.com/watch?v=pVPOdj9iZ5s&t=10s>

Gas turbine



Gas Turbine Manufacturers

1. ABB
2. Alstom/Ansaldo
3. General Electric
4. Mitsubishi
5. Siemens
6. Bharat Heavy Electricals
6. Westinghouse

Brayton Cycle analysis

$$Q_{in} = c_p(T_3 - T_2) \text{ and } Q_{out} = c_p(T_4 - T_1)$$

$$\text{Efficiency} = \frac{Q_{in} - Q_{out}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$$

$$= 1 - \frac{c_p(T_4 - T_1)}{c_p(T_3 - T_2)}$$

$$\eta = 1 - \frac{T_1}{T_2} \left[\frac{\frac{T_4}{T_1} - 1}{\frac{T_3}{T_2} - 1} \right]$$

We have, $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$, $\frac{T_2}{T_1} = \frac{P_2 V_2}{P_1 V_1}$

Also $P_1 V_1^k = P_2 V_2^k$, $k = c_p / c_v$, $V_2 / V_1 = (P_1 / P_2)^{1/k}$

Thus $T_2 / T_1 = (P_2 / P_1)^{(k-1)/k}$,

Similarly

$$T_4 / T_3 = (P_4 / P_3)^{(k-1)/k} = (P_1 / P_2)^{(k-1)/k}$$

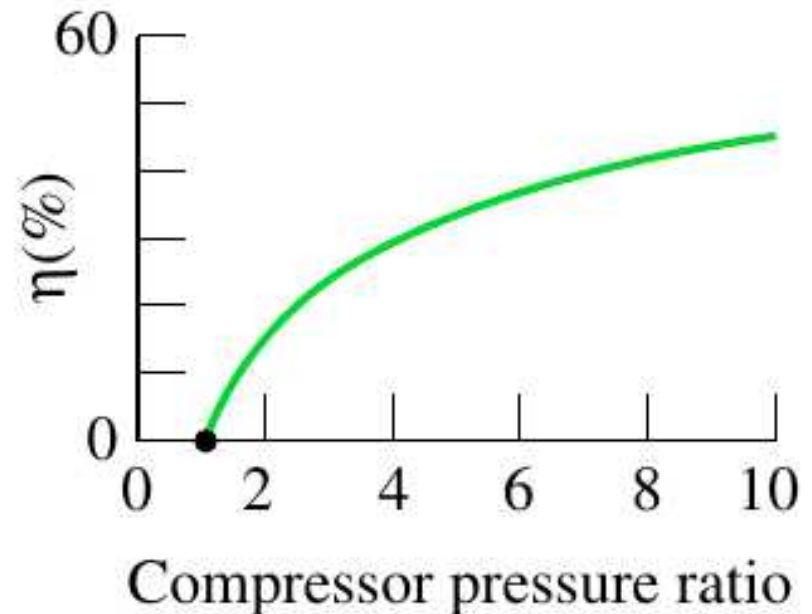
Brayton Cycle analysis

We have $\frac{T_2}{T_1} = \frac{T_3}{T_4}, \frac{T_3}{T_2} = \frac{T_4}{T_1}$

$$\eta = 1 - \frac{T_1}{T_2} \left[\frac{\frac{T_4}{T_1} - 1}{\frac{T_3}{T_2} - 1} \right]$$

$$\text{Efficiency} = 1 - T_1/T_2$$

$$\text{Efficiency} = 1 - 1/(P_2/P_1)^{(k-1)/k}$$



Compressor Pressure Ratio for Maximum Net Work

Net work = Turbine work – Compressor work

$$\begin{aligned}W_{\text{net}}/m &= (h_3 - h_4) - (h_2 - h_1) \\&= c_p [(T_3 - T_4) - (T_2 - T_1)] \\&= c_p T_1 \left[\frac{T_3}{T_1} - \frac{T_4}{T_3} \frac{T_3}{T_1} - \frac{T_2}{T_1} + 1 \right]\end{aligned}$$

We have $T_2/T_1 = (P_2/P_1)^{(k-1)/k}$ and $T_4/T_3 = (P_1/P_2)^{(k-1)/k}$

$$W_{\text{net}}/m = c_p T_1 \left[\frac{T_3}{T_1} - \frac{T_3}{T_1} \left(\frac{P_1}{P_2} \right)^{\left[\frac{k-1}{k} \right]} - \left(\frac{P_2}{P_1} \right)^{\left[\frac{k-1}{k} \right]} + 1 \right]$$

For specified values of T_1 , T_3 and c_p differentiating w.r.t

$$(P_2/P_1)$$

Compressor Pressure Ratio for Maximum Net Work

$$\begin{aligned}\frac{\partial(W/m)}{\partial(P_2/P_1)} &= c_p T_1 \left[\frac{(k-1)}{k} \left(\frac{T_3}{T_1} \left(\frac{P_2}{P_1} \right)^{\left[\frac{-(k-1)}{k} - 1 \right]} \right) - \left[\frac{(k-1)}{k} \right] \left(\frac{P_2}{P_1} \right)^{\left[\frac{k-1}{k} \right] - 1} \right] \\ &= c_p T_1 \left(\frac{(k-1)}{k} \right) \left(\frac{T_3}{T_1} \left[\frac{P_2}{P_1} \right]^{\left[\frac{(-2k+1)}{k} \right]} - \left[\frac{P_2}{P_1} \right]^{\left(\frac{-1}{k} \right)} \right)\end{aligned}$$

Equating to zero

$$\left(\frac{T_3}{T_1} \right) \left[\frac{P_2}{P_1} \right]^{\left[\frac{(-2k+1)}{k} \right]} - \left[\frac{P_2}{P_1} \right]^{\left(\frac{-1}{k} \right)} = 0$$

$$\left(\frac{T_3}{T_1} \right) \left[\frac{P_2}{P_1} \right]^{\left[\frac{(-2k+1)}{k} \right]} = \left[\frac{P_2}{P_1} \right]^{\left(\frac{-1}{k} \right)}$$

$$\left(\frac{T_3}{T_1} \right) = \left[\frac{P_2}{P_1} \right]^{\left[\frac{(-1+2k-1)}{k} \right]} = \left(\frac{T_3}{T_1} \right) = \left[\frac{P_2}{P_1} \right]^{\left[\frac{2(k-1)}{k} \right]}$$

$$\left(\frac{P_2}{P_1} \right) = \left[\frac{T_3}{T_1} \right]^{\left[\frac{k}{2(k-1)} \right]}$$

Compressor Pressure Ratio for Maximum Net Work

We have

$$W_{\text{net}}/m = c_p T_1 \left[\frac{T_3}{T_1} - \frac{T_3}{T_1} \left(\frac{P_1}{P_2} \right)^{\left[\frac{k-1}{k} \right]} - \left(\frac{P_2}{P_1} \right)^{\left[\frac{k-1}{k} \right]} + 1 \right]$$

Substituting $\left(\frac{P_2}{P_1} \right) = \left[\frac{T_3}{T_1} \right]^{\left[\frac{k}{2(k-1)} \right]}$

$$c_p T_1 \left[\frac{T_3}{T_1} - \frac{T_3}{T_1} \left(\left[\frac{T_3}{T_1} \right]^{\left[\frac{-k}{2(k-1)} \right]} \right)^{\left[\frac{k-1}{k} \right]} - \left[\frac{T_3}{T_1} \right]^{\left[\frac{k}{2(k-1)} \right]} \left[\frac{k-1}{k} \right] + 1 \right]$$

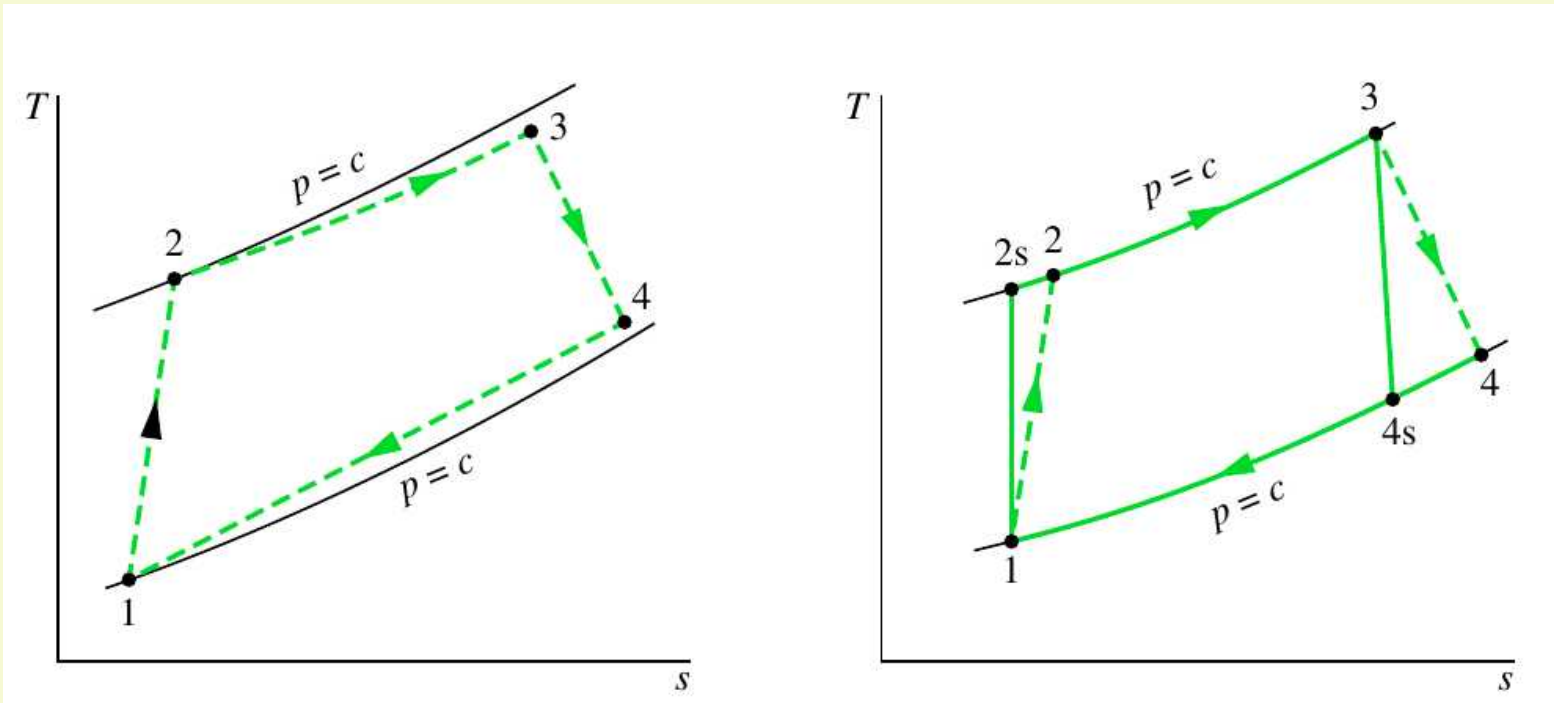
Simplifying $c_p T_1 \left[\frac{T_3}{T_1} - \frac{T_3}{T_1} \left[\frac{T_3}{T_1} \right]^{-\frac{1}{2}} - \left[\frac{T_3}{T_1} \right]^{\frac{1}{2}} + 1 \right]$

$$W_{\text{net}} = c_p T_1 \left[\frac{T_3}{T_1} - \frac{T_3}{T_1}^{\frac{1}{2}} - \frac{T_3}{T_1}^{\frac{1}{2}} + 1 \right]$$

$$W_{\text{net}} = c_p T_1 \left[\frac{T_3}{T_1} - 2 \frac{T_3}{T_1}^{\frac{1}{2}} + 1 \right]$$

$$W_{\text{net}} = c_p \left[T_3 - 2 T_3 T_1^{\frac{1}{2}} + T_1 \right] = c_p \left[\sqrt{T_3} - \sqrt{T_1} \right]^2$$

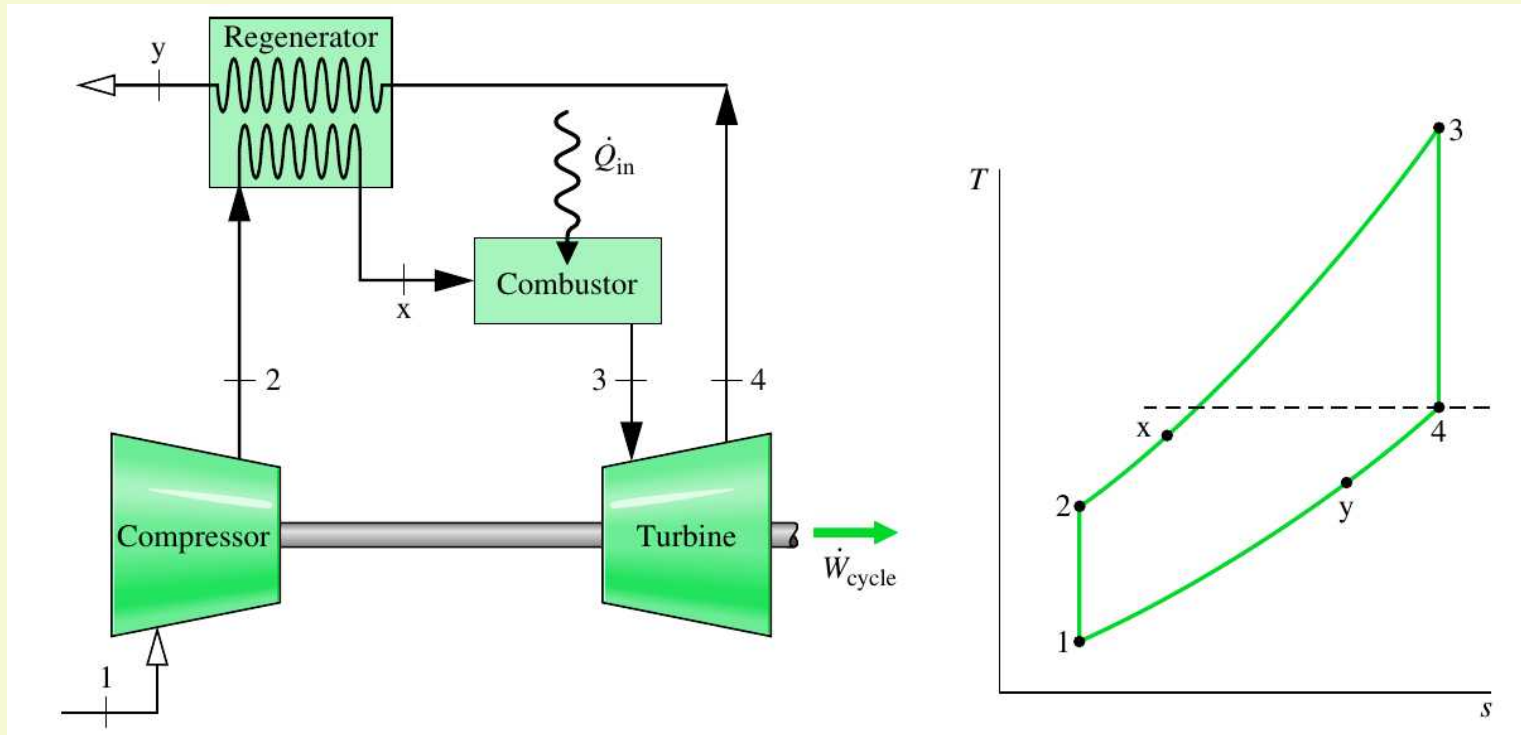
Gas Turbine irreversibilities



$$\eta_t = \frac{(W_t/m)}{(W_t/m)_s} = \frac{h_3 - h_4}{h_3 - h_{4s}} = \frac{T_3 - T_4}{T_3 - T_{4s}}$$

$$\eta_c = \frac{(W_c/m)_s}{(W_c/m)} = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{T_{2s} - T_1}{T_2 - T_1}$$

Regenerative Gas Turbines

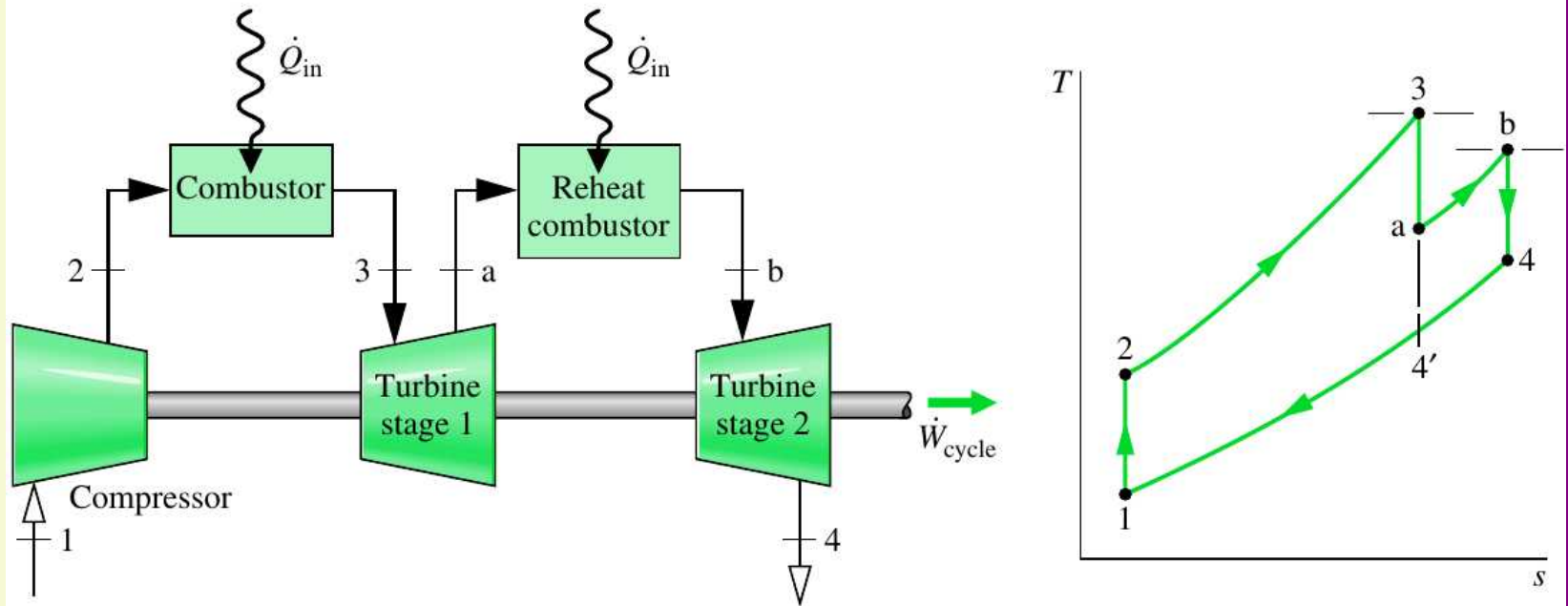


Net work per unit mass increases

Thermal efficiency increases

$$\eta_{reg} = \frac{h_x - h_2}{h_4 - h_2}$$

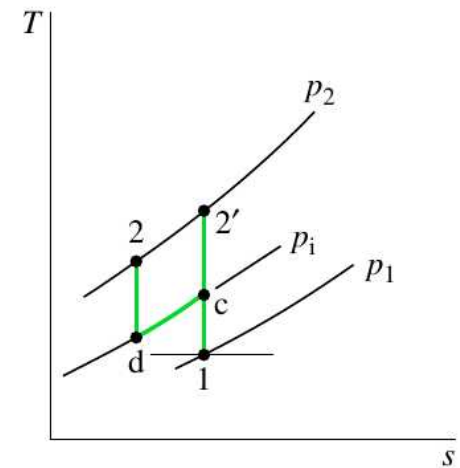
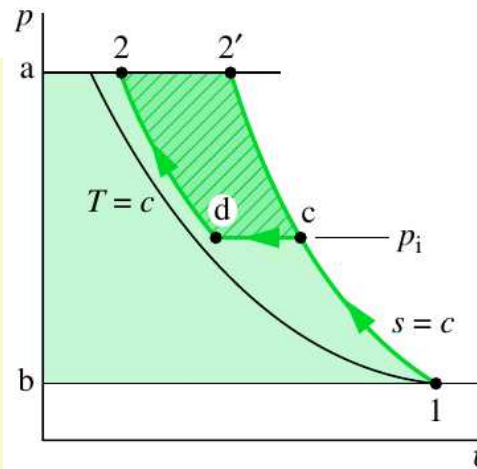
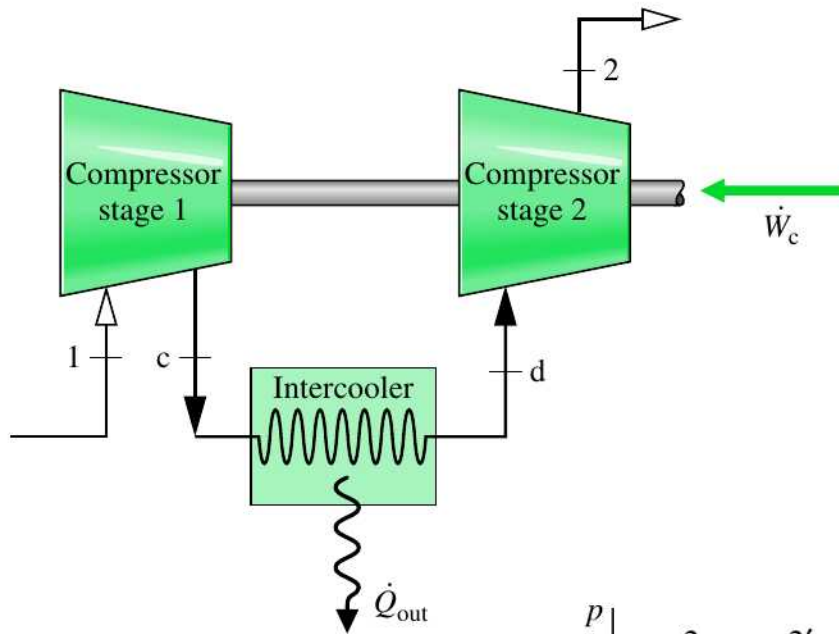
Gas Turbines with reheat



Net work per unit mass increases

Thermal efficiency may not increase

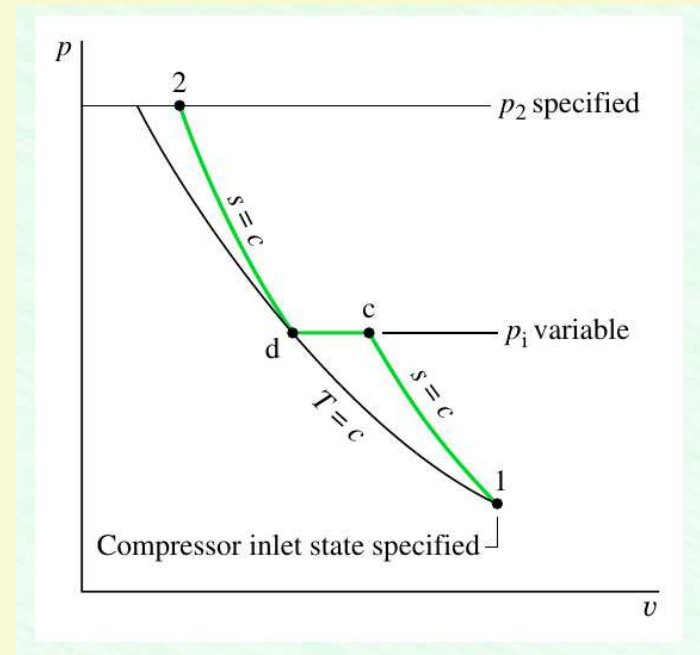
Compressor with intercooling



Net work per unit mass decreases

Intercooler pressure for minimum Compressor Work

Compressor work -minimum at constant temperature



$$\frac{W_c}{m} = (h_c - h_1) + (h_2 - h_d)$$

Assuming temperature $T_1 = T_d$ after inter-cooling

$$\begin{aligned} \frac{W_c}{m} &= c_p [(T_c - T_1) + (T_2 - T_1)] \\ &= c_p T_1 \left[\frac{T_c}{T_1} + \frac{T_2}{T_1} - 2 \right] \end{aligned}$$

Intercooler pressure for minimum Compressor Work

We have $\frac{T_c}{T_1} = \left(\frac{P_i}{P_1}\right)^{\left[\frac{k-1}{k}\right]}$ and $\frac{T_2}{T_d} = \left(\frac{P_2}{P_i}\right)^{\left[\frac{k-1}{k}\right]}$

Substituting $\frac{W_c}{m} = c_p T_1 \left[\left(\frac{P_i}{P_1}\right)^{\left[\frac{k-1}{k}\right]} + \left(\frac{P_2}{P_i}\right)^{\left[\frac{k-1}{k}\right]} - 2 \right]$

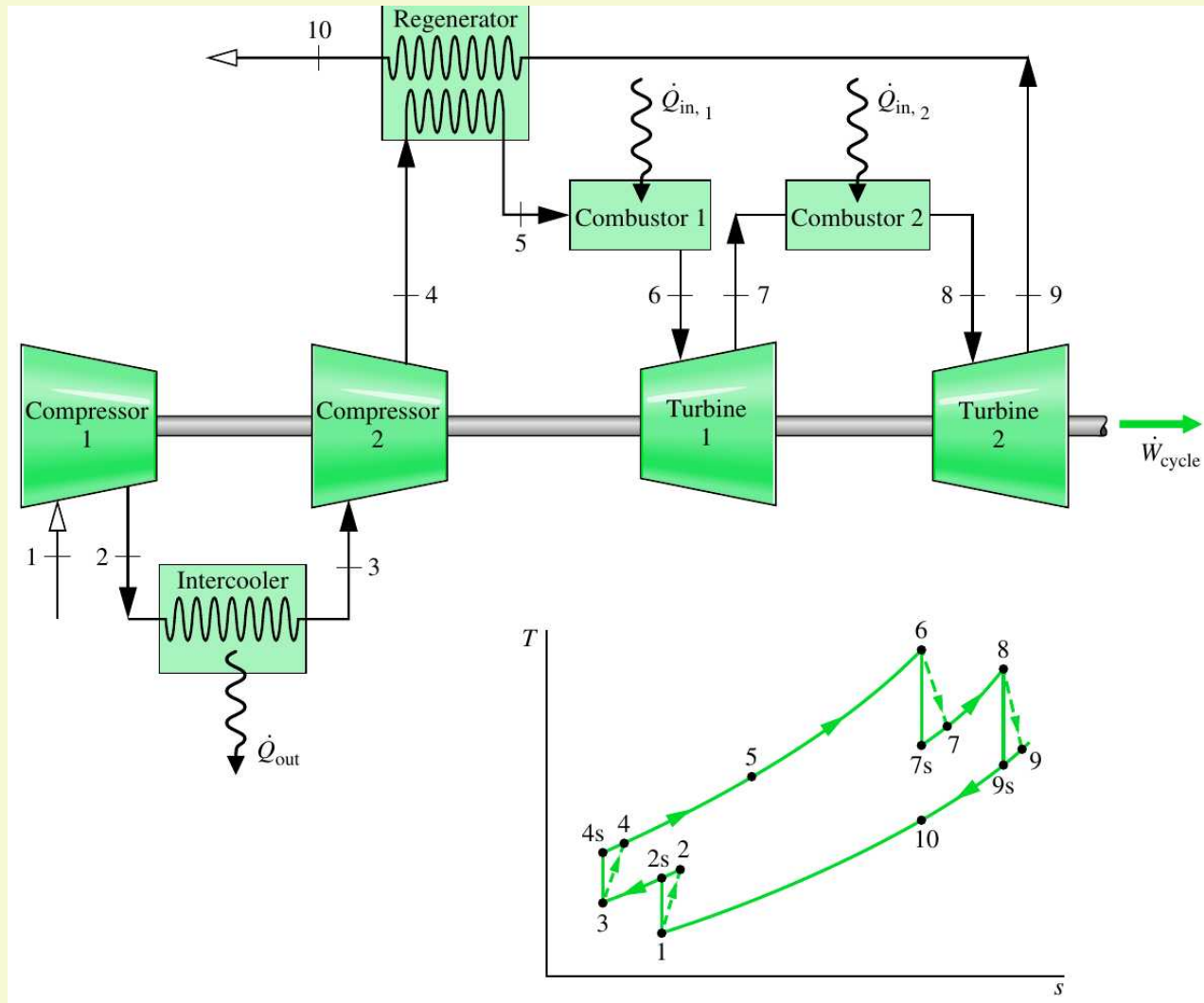
Optimum P_i $\frac{\partial\left(\frac{W_c}{m}\right)}{\partial p_i} = \frac{\partial\left(c_p T_1 \left[\left(\frac{P_i}{P_1}\right)^{\left[\frac{k-1}{k}\right]} + \left(\frac{P_2}{P_i}\right)^{\left[\frac{k-1}{k}\right]} - 2 \right]\right)}{\partial p_i}$

$$= c_p T_1 \left(\frac{k-1}{k}\right) \left[\left(\frac{P_i}{P_1}\right)^{\left[\frac{-1}{k}\right]} \left(\frac{1}{P_1}\right) + \left(\frac{P_2}{P_i}\right)^{\left[\frac{-1}{k}\right]} \left(\frac{-P_2}{P_i^2}\right) \right]$$
$$= \frac{c_p T_1}{P_i} \left(\frac{k-1}{k}\right) \left[\left(\frac{P_i}{P_1}\right)^{\left[\frac{k-1}{k}\right]} - \left(\frac{P_2}{P_i}\right)^{\left[\frac{k-1}{k}\right]} \right]$$

Equating to Zero

$$P_i = \sqrt{P_1 P_2}$$

Regenerative Gas Turbines with reheat and intercooling



Net work per unit mass increases

Overall Thermal efficiency increase

Combustion Process

Combustion Steps

- Mixing of a fine spray of droplets with air
- Vaporization of the droplets
- Breaking down of heavy hydrocarbons into lighter fractions
- Intimate mixing with oxygen molecules
- Finally the chemical reactions - combustion
- Uses 100:1 Air Fuel ratio

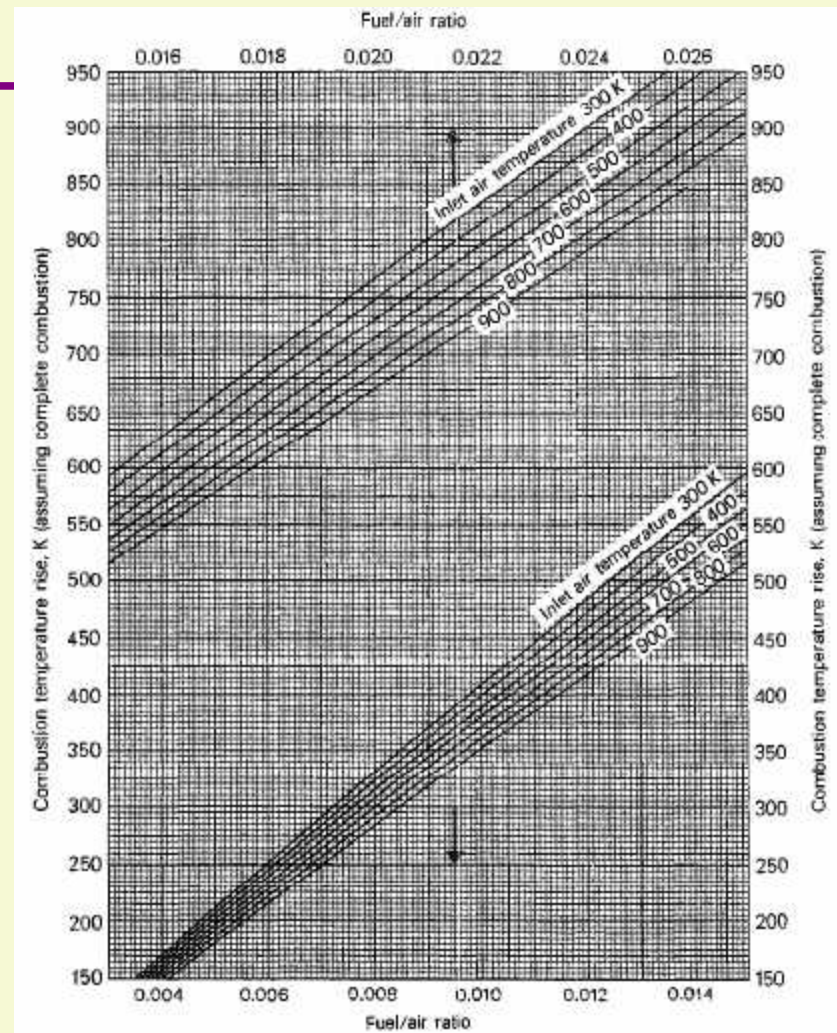
Combustion chamber performance

Factors affecting the performance

- Pressure loss
 - Friction and turbulence
 - Rise in temperature due to combustion.
- Combustion efficiency
- Outlet temperature distribution
- Stability limits
- Combustion intensity.
- $CI \text{ (kW/m}^3\text{atm)} = \frac{\text{Heat release rate}}{\text{Comb. Vol x Pressure}}$

Lower the value of the combustion intensity the easier it is to design a combustion system which will meet all the desired requirements.

Combustion efficiency

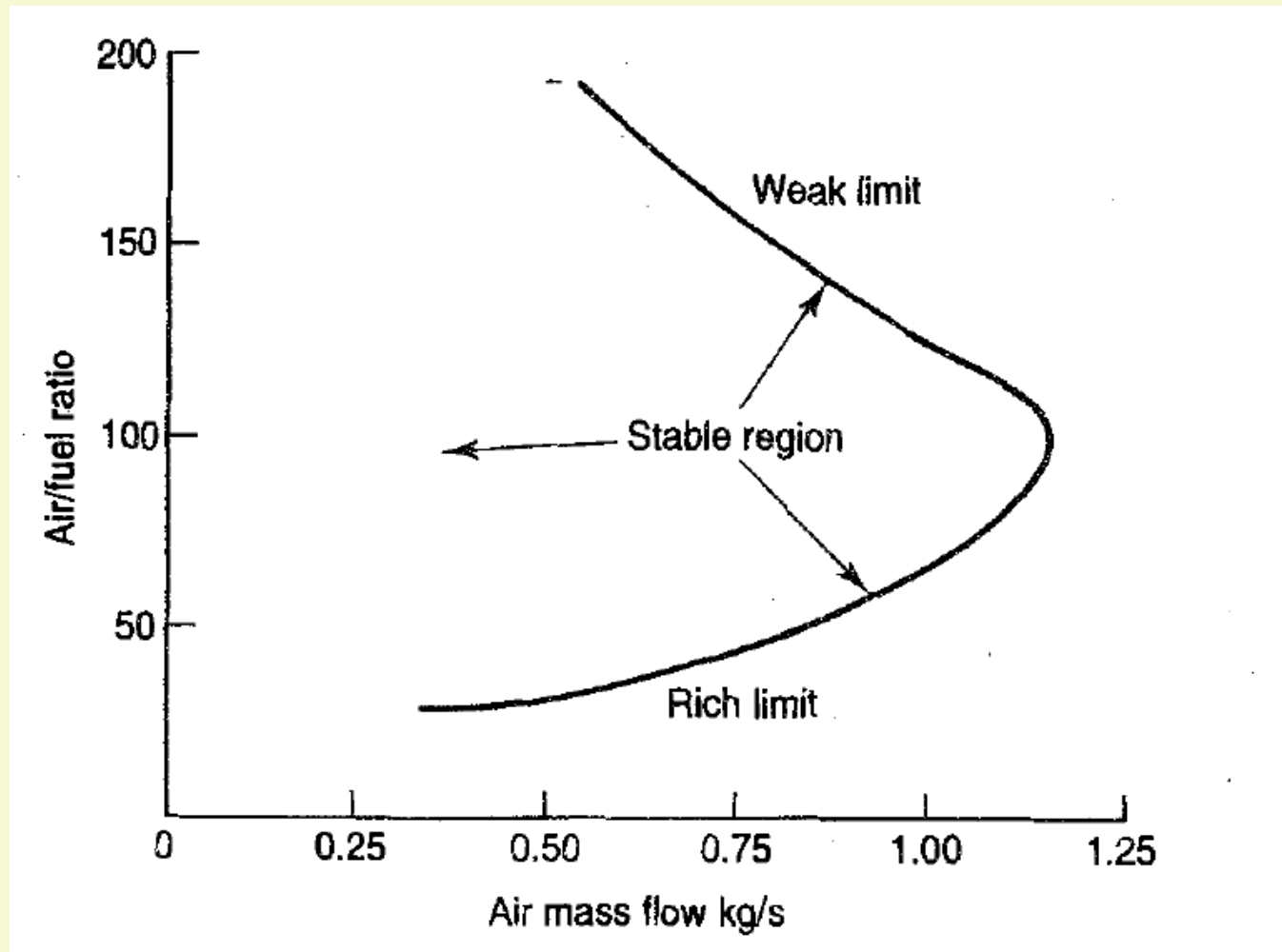


Fuel air ratio = f

Combustion efficiency = $\frac{\text{Theoretical } f \text{ for given } \Delta T}{\text{Actual } f \text{ for given } \Delta T}$

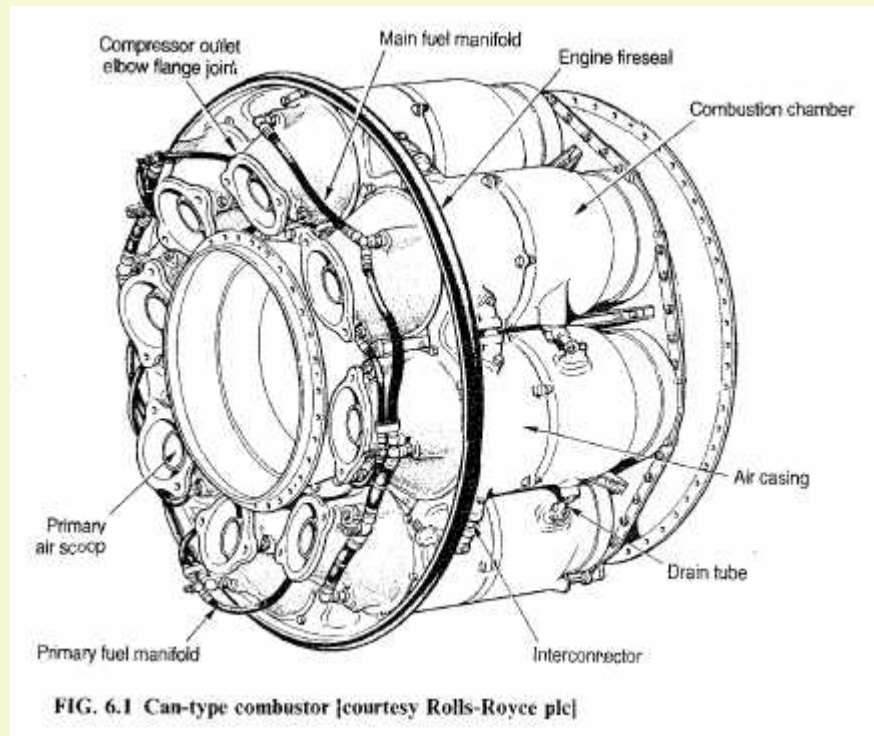
Specific fuel consumption = f/W_{net}

Combustion stability loop



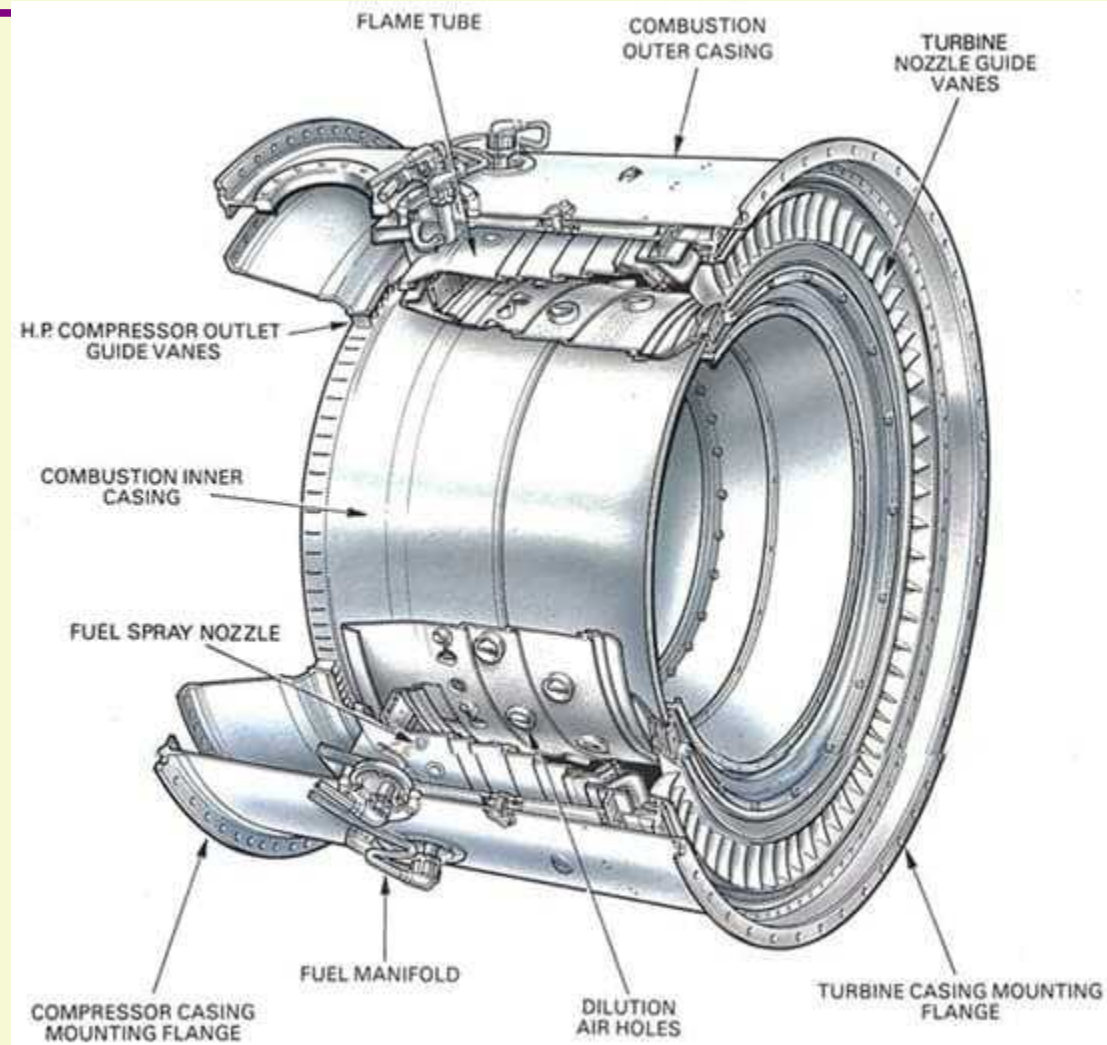
Combustion stability loop

Combustion



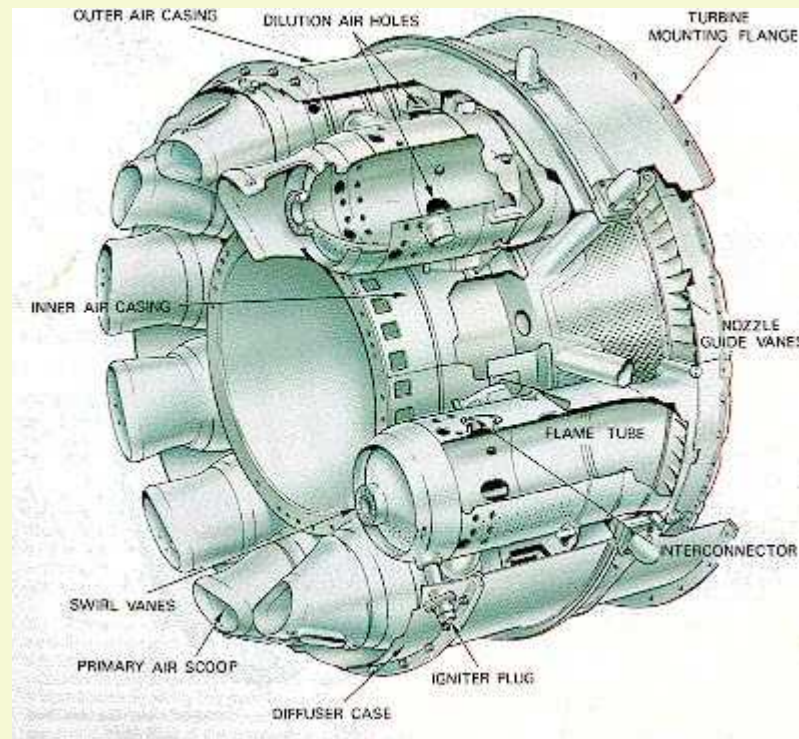
Can type combustor

Combustion



Annular type combustor

Combustion



Cannular type combustor

Gas Turbine Power plant

Advantages	Disadvantages
It is simple to design & construct compared to a Steam Power station, since no boilers are required.	Problem of starting. Compressor needs to be operating thus external source of power is necessary.
Much smaller compared to Steam Power station of same capacity	Net output is low since greater power is used to drive compressor.
Lower operating costs	Overall efficiency of plant is low $\approx 20\%$ because of exhaust gases still containing heat. only efficient in a Combined cycle configuration
Less water used since there is no need for a condenser	Temperature of combustion chamber is too high thus resulting in a lower life.
Maintenance charges are low	
Can be started quickly	
No stand-by losses	

GAS TURBINE COMBINED CYCLE Power Plant System Schematic

